

HEALTH SYNAPSE

Intelligent Healthcare Resource Connectivity Platform



Presented By Hexora Team

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PROBLEM STATEMENT

Student Innovation-Cutting-edge technology in these sectors continues to be in demand. Recent shifts in healthcare trends, growing populations also present an array of opportunities for innovation

SOLUTION:

A one-stop smart healthcare platform that provides quick access to hospitals, doctors, emergency services, appointments, blood and bed availability, and healthcare assistance.

GOAL:

To reduce healthcare access time and improve patient support by bringing essential medical services and information together in one user-friendly platform.

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Strengths

- Multiple healthcare services in one platform
- Quick hospital and doctor discovery
- Emergency support feature
- Blood and bed availability information
- Appointment and donation booking

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Weaknesses

- Requires regular data updates
- Current prototype may use limited/static data
- Requires internet connectivity
- Chatbot has limitations in providing healthcare guidance

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Opportunities

- Real-time hospital data integration
- AI-powered healthcare assistance
- GPS-based nearby hospital search
- Mobile application development
- Integration with more hospitals and healthcare services

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Threats

- Outdated or incorrect healthcare information
- Data privacy and security concerns
- Dependence on hospital participation
- Technical failures during emergencies
- User trust and adoption challenges

KEY FEATURES / MODULES



Hospital Discovery



Doctor Search & Schedules



Appointment Booking



Emergency / Ambulance Support



Blood Availability



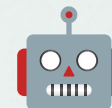
Bed Availability



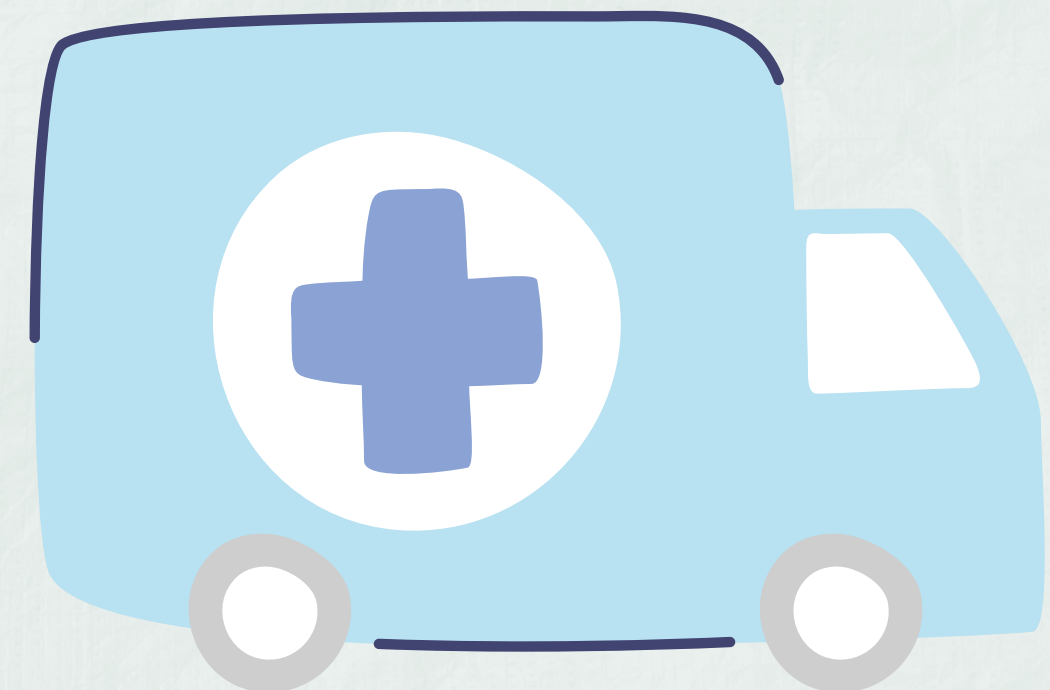
Blood Donation Scheduling



Veterinary Hospitals



TAN AI Chatbot



DATA ARCHITECTURE & API FEASIBILITY



Blood Supply:

Tech Sourcing: Fetch live government data via the e-RaktKosh API, blended with a local Firebase-managed student/volunteer donor pool.



Hospital Beds & Oxygen:

Tech Sourcing: Use modern FHIR / HL7 medical data standards to link directly with open-source Hospital Management Information Systems (HMIS).



Ambulances:

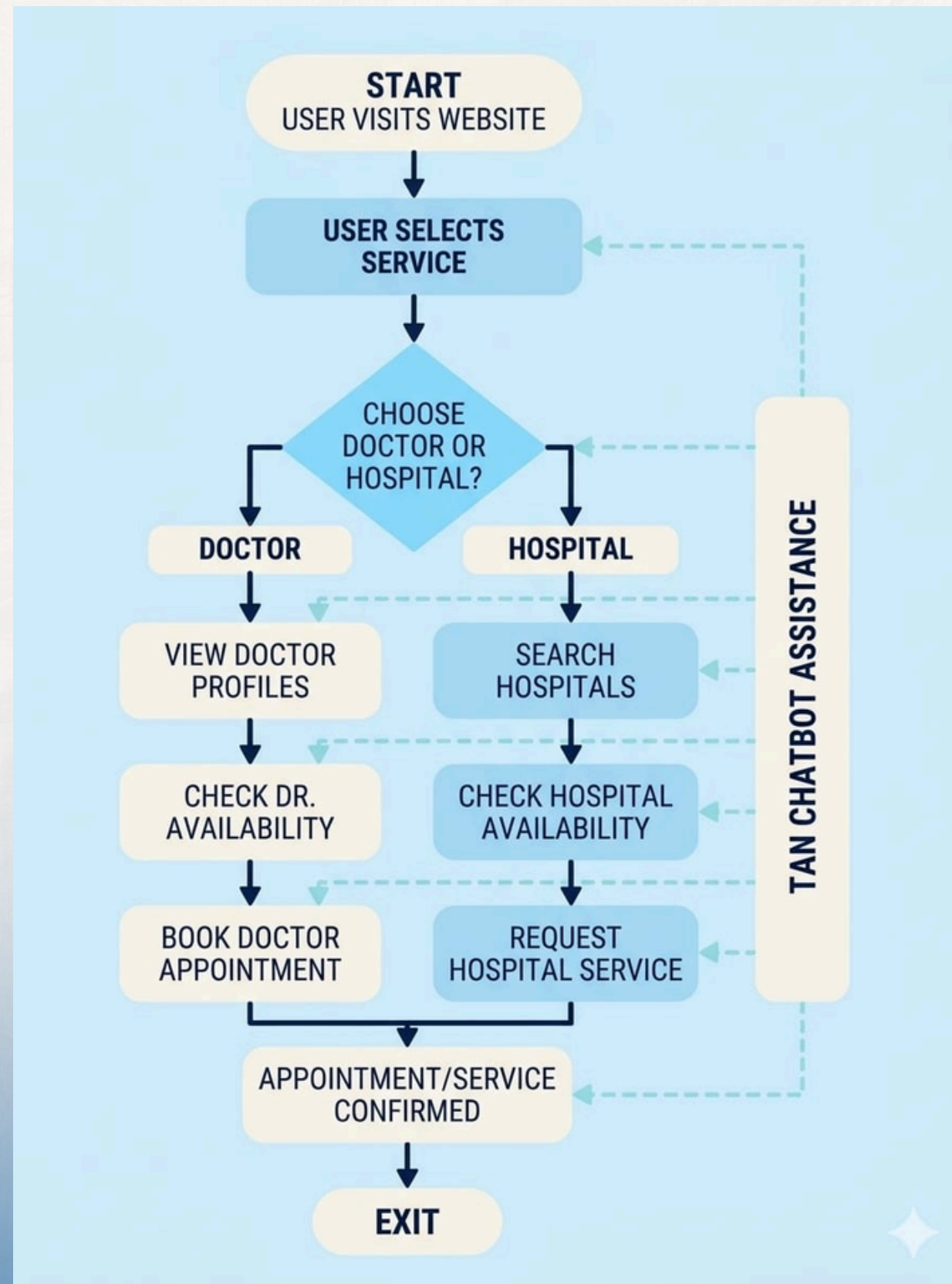
Tech Sourcing: Built-in progressive location-tracking using WebSockets (for live, low-latency map movements) rather than heavy HTTP polling to save mobile battery.



Doctor Appointments:

Tech Sourcing: A unified queuing algorithm that syncs with real-time OPD schedules, allowing automated push notifications if a doctor is running late.

Workflow



RISK MITIGATION

- Outdated Data → Regular updates
- Data Privacy → Secure data handling
- Chatbot Errors → Professional consultation guidance
- API Failure → Backup data
- Emergency Risk → Direct emergency support

IMPACT, SOCIAL BENEFITS & SCALABILITY

40% Reduction

In Emergency
Inter-Hospital
Transit Time

100% Inclusive

Voice-Enabled for
Illiterate & Rural
Citizens

24/7 Monitoring

Continuous High-
Risk Vital Anomaly
Escalation

MULTI-STAKEHOLDER BENEFITS

- Patients: Easy access to hospitals, doctors & appointments.
- Emergency Users: Quickly find hospitals with available resources.
- Hospitals: Better bed management & reduced overcrowding.
- Blood Banks: Improved blood availability through shared listings.

NATIONAL SCALE & GOVT ALIGNMENT

- ABDM Integration: Supports digital healthcare connectivity.
- District-Level Expansion: Can be deployed across different regions.
- Offline Support: Useful during limited internet connectivity.
- Scalable Platform: Can integrate more hospitals and healthcare services.



RESEARCH – EXISTING WEBSITE: PRACTO

- Practo provides doctor search, appointments, online consultations, lab tests, medicines and health records.
- Gap identified: Emergency resource discovery, real-time bed/blood availability and blood-donation services are not the central focus of the platform.
- Our approach: Smart Health Assistant combines these healthcare-access services into one platform.

REFERENCE

https://www.practo.com/?utm_source=chatgpt.com

THANK
YOU